

Money and Success in Football

DOES FINANCIAL STRENGTH PREDICT LEAGUE TITLES?

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Does Financial Strength Predict League Position?

Research Question: Does a squad's financial rank within its league systematically determine where it finishes in the final standings?

Why this matters:

- Europe's five major leagues generate over **€18 billion** in annual revenue; transfer spending exceeded **€12 billion** in 2023–24 alone.
- UEFA introduced **Financial Fair Play** precisely to limit financial dominance — implying money *does* buy success.
- If outcomes are financially predetermined, competitive drama is fundamentally reduced for fans of smaller clubs.

This study uses 976 team-seasons from 5 leagues and 10 seasons (2015–16 to 2024–25) to answer with evidence, not opinion.

Two contrasting cases — one financially expected, one statistically miraculous — motivate why the question requires careful empirical investigation.

Two Cases: One Expected, One Miraculous

Leicester City 2015–16

Financial Rank	11th of 20
League	Premier League
Squad value	£161M (avg: £458M)
Model prediction	11.5th place
Actual finish	1st (Champion)
Residual	–10.5 positions
Bookmaker odds	5 000-to-1

The largest single-season overperformance in the entire 10-year dataset — a statistical shock that occurs with probability well below 0.1%.

Bayer Leverkusen 2023–24

Financial Rank	2nd (Bundesliga)
Result	Bundesliga champions
Historic feat	First unbeaten Bundesliga season
Context	Ended Bayern's 11-year streak
Residual	–1.8 (within normal range)

Remarkable by football standards, but statistically *expected*: the second-richest squad winning is precisely what the financial hierarchy predicts.

These two cases bracket the full spectrum of outcomes: from the financially-expected champion to the greatest financial upset in modern football.

Data and Scope: 976 Team-Seasons Across Five Leagues

Two data sources:

- **ESPN FC** — final league standings, all five leagues, seasons 2015–16 through 2024–25.
- **Transfermarkt** — squad total market value (€M) at the start of each season.

League	N	Mean MV	$r(\text{FR}, \text{Pos})$
Bundesliga	180	€237M	0.783
La Liga	200	€257M	0.787
Ligue 1	196	€185M	0.791
Premier League	200	€458M	0.801
Serie A	200	€246M	0.890
All	976	€278M	0.813

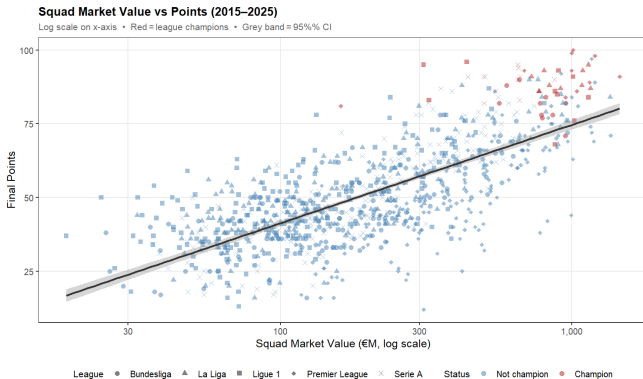
All r values: $p < 0.001$.

Key Variable: Financial Rank (FR)

- Squads ranked by market value **within each league-season**: FR = 1 richest, FR = n least wealthy.
- **Why rank, not raw euros?** A €300M squad is the *richest* in Ligue 1 but only the *fifth-richest* in the Premier League. Ranking within leagues removes this scale problem.
- FR = 1 always means “the wealthiest club in this league this season,” regardless of whether that squad is worth €600M or €150M.

Even before any model: correlations above 0.8 tell us money is a very powerful predictor. Pearson $r = 0.813$, Spearman $\rho = 0.814$; both $p < 0.001$, $N = 976$.

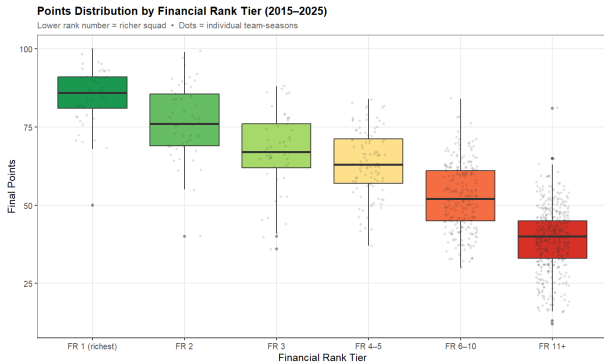
Before Ranking: Does Raw Squad Value Predict Points?



- Log-scale x-axis: squad value spans €20M to €1.3B.
- **Red dots** = league champions — clustered firmly in the upper-right.
- The relationship is strong, but raw euros mix two different things: a club's wealth, and its league's overall wealth.

A €300M squad is rich in Ligue 1 but mid-table in the Premier League. Raw value cannot be compared across leagues — this is exactly why we built Financial Rank next.

Financial Rank Creates a Balanced, Meaningful Hierarchy



Why Uniformity Matters

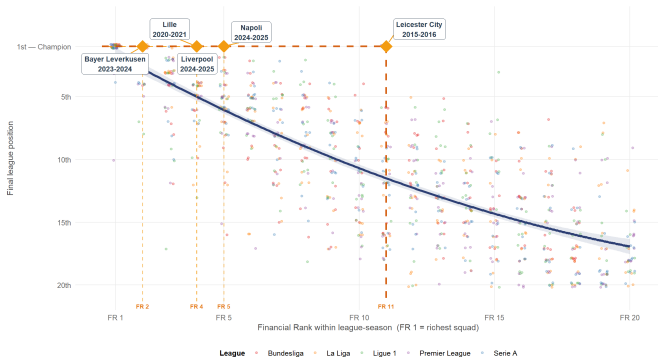
- Every league-season contains exactly as many clubs as rank positions, so each FR value appears equally often by construction.
- FR = 1 appears exactly 50 times (once per league-season), as do FR = 2, FR = 3, and so on.
- No structural concentration at any particular rank: the system is **clean and balanced**.

The boxplot confirms a monotonic decline in league points as Financial Rank increases — the ranking system is balanced and meaningfully stratifies performance.

Richer Squads Finish Higher — Across All 976 Team-Seasons

Richer Squads Finish Higher — With Notable Exceptions

N = 976 team-seasons • Pearson $r = 0.81$, $p < 0.001$ • Model H quadratic fit • Gold = key story clubs



Financial Rank normalised within each league-season. Axis reversed: top = better.

$$r = 0.813 \quad R^2 = 0.671$$

Pearson correlation
and variance explained

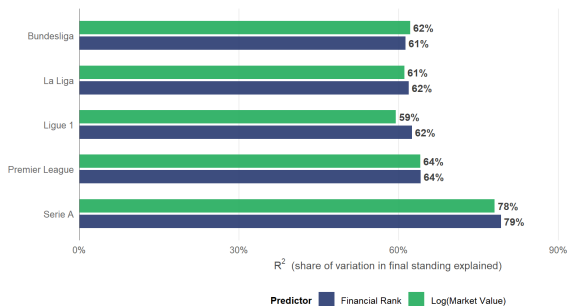
FR vs. Final Position,
 $N = 976$, $p < 0.001$

- FR = 1–2 clubs cluster strongly near the top of their league.
- $FR \geq 15$ clubs cluster near the bottom.
- The fitted curve (Model H) bends at the bottom: diminishing returns.

The Financial Hierarchy Holds in Every League

Money Explains 61–79% of Final League Position

R^2 per league • Both predictors significant ($p < 0.001$) in all leagues

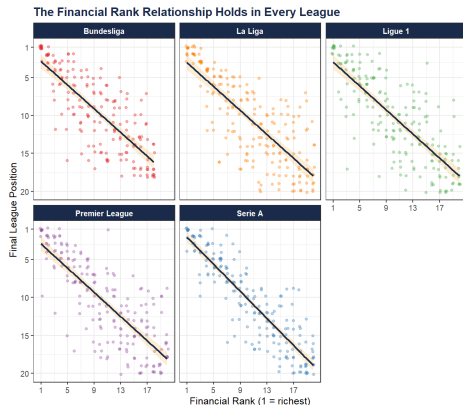


OLS: League_Position □ predictor, estimated separately per league. Serie A shows the strongest relationship ($R^2 = 0.79$).

League	r	R^2
Bundesliga	0.783	0.613
La Liga	0.787	0.619
Ligue 1	0.791	0.625
Premier League	0.801	0.641
Serie A	0.890	0.792
All (pooled)	0.813	0.671

Financial Rank alone explains 61–79% of final position *within each individual league*. Serie A shows the strongest relationship ($R^2 = 0.79$); even the weakest (Bundesliga, $R^2 = 0.61$) is very high by any academic standard.

The Pattern Holds in Every League



Five leagues, one pattern. The slope is statistically indistinguishable across Bundesliga, La Liga, Ligue 1, Premier League, and Serie A — confirming the financial hierarchy is universal, not a quirk of one competition. Serie A shows the tightest fit; even the loosest (Ligue 1) is extremely strong by any standard.

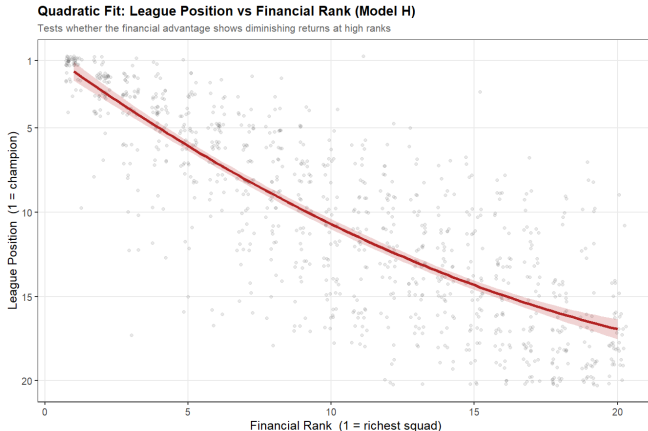
Model Selection: AIC Identifies Model H as Best Specification

Mdl	Formula	R^2	Adj. R^2	AIC
A	FR only (linear baseline)	0.661	0.660	5104.4
C	Log(Market Value) only	0.533	0.533	5415.5
H	FR + FR² (quadratic)	0.671	0.671	5075.1
H4	FR + FR ² + League + Season	0.672	0.669	5083.9
J	Log(FR)	0.626	0.626	5199.0
E	Champion ~ FR (Logit)	McFadden = 0.548		182.4

Model H wins on AIC: adding one extra term (FR²) lowers AIC by 29.3 points vs. the linear baseline. Extended Model H4 (league + season controls) scores higher AIC than H — the controls add no explanatory power ($F(5) = 0.253$, $p = 0.939$).

OLS response: League Position (1 = champion), $N = 976$. Lower AIC = better fit per parameter. Bold = preferred specification. Model E uses logistic regression for the binary outcome (champion vs. not).

Model H: Diminishing Returns to Financial Advantage



Model H Equation

$$\widehat{\text{Pos}} = 0.42 + 1.23 \text{FR} - 0.020 \text{FR}^2$$

- $\hat{\beta}_1 = 1.231$ (SE 0.076),
 $p < 0.001$
- $\hat{\beta}_2 = -0.020$ (SE 0.004),
 $p < 0.001$
- $R^2 = 0.671$

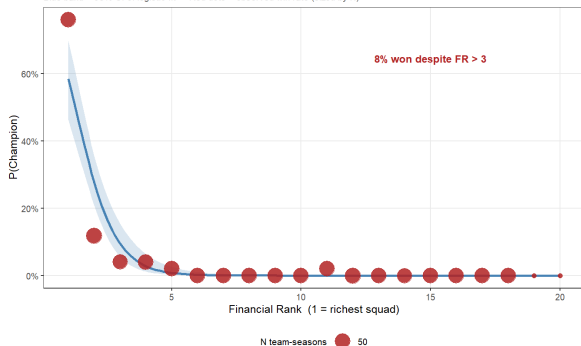
Marginal cost of one rank step

- FR 1 $\rightarrow$ 2: $\approx$ **1.19** positions lost
- FR 15 $\rightarrow$ 16: $\approx$ **0.62** positions lost
- Negative quadratic confirmed:
 $F = 31.73$, $p < 0.001$

Championship Probability Plummet With Financial Rank

Predicted Probability of Championship vs Financial Rank (Model E)

Blue band = 95% CI of logistic fit • Red dots = observed win rate (sized by N)



FR	$P(\text{Champion})$
1 (richest)	58.5 %
2	27.8 %
3	9.5 %
5	0.8 %
11	<0.01 %

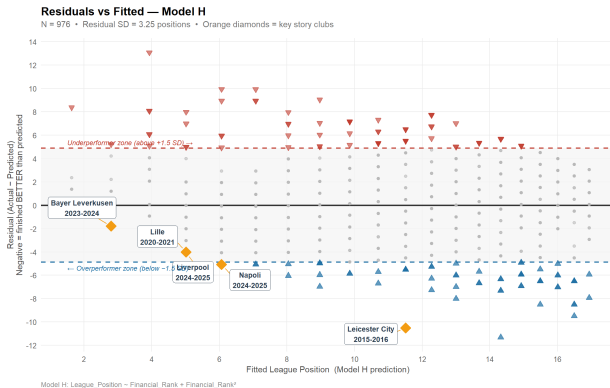
Model E: $\text{Champion} \sim \text{FR}$ (logistic).

$\hat{\beta}_{\text{FR}} = -1.298$, $p < 0.001$. Each rank step multiplies odds by 0.273 (−72.7%).

McFadden $R^2 = 0.548$.

FR = 1: 58.5 % title probability — $\approx 3\times$ the baseline for a 20-team league. This is structural advantage, not certainty: winning from FR = 11 is virtually impossible by this model.

Who Beat the Model? Residual Analysis



What Is a Residual?

Residual = Actual - Predicted

- **Negative:** finished **better** than predicted (**overperformer**)
- **Positive:** finished **worse** than predicted (**underperformer**)
- Near zero: correctly predicted

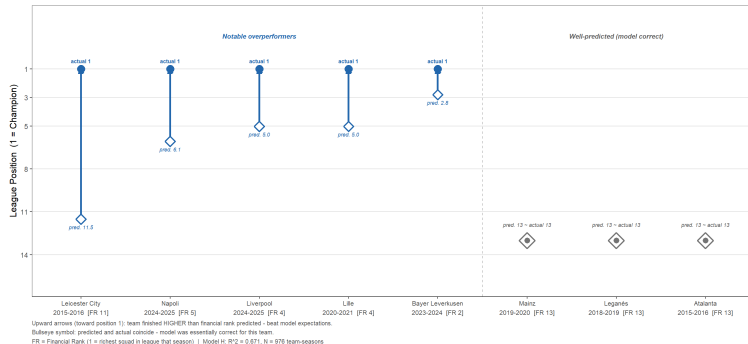
Dashed lines mark $\pm 1.5\text{ SD}$ ($\approx \pm 4.9$ positions). Teams outside this band are genuine statistical outliers — performance unexplained by finances alone.

Five Teams That Defied Their Financial Rank

Predicted vs Actual League Position: Selected Team-Seasons (Model H)

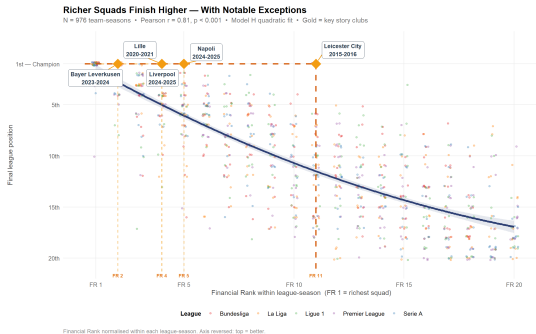
Diamond = Model H predicted position (FR + FR²) · Circle = actual final league position · Bullseye = predicted matches actual

◆ Outperformed: actual finish better than Model H prediction



Each arrow spans the gap between what financial rank predicted and what happened on the pitch. A grey bullseye means the model was correct; a blue arrow pointing up means the team beat its financial expectation. Leicester City's arrow spans 10.5 positions — the largest single gap in all 976 team-seasons.

Leicester City 2015–16: The Greatest Statistical Upset



Leicester City 2015–16

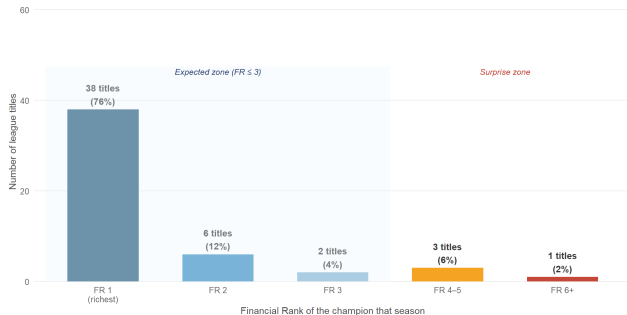
Financial Rank	11th of 20 (PL)
Predicted pos.	11.5th
Actual pos.	1st (Champion)
Residual	−10.5 positions
SD units	−3.2 SD
Cook's Distance	Highest in dataset

4 of 50 champions (8 %) came from $FR > 3$. Napoli (FR5), Lille (FR4), Liverpool (FR4) were within 1 SD of expectation. Only Leicester (FR = 11) is a genuine statistical outlier.

How Often Does Money Actually Win the Title?

76% of Titles Were Won by the League's Richest Club

50 champions over 10 seasons across 5 leagues • 92% came from the financial top 3



FR 6+ = 4 champions: Leicester City (FR 11, 2015–16), Napoli (FR 5, 2024–25), Lille (FR 4, 2020–21), Liverpool (FR 4, 2024–25).

- **38 of 50 titles (76%)** went to the single richest club in the league that season.
- **46 of 50 titles (92%)** came from the financial top 3.
- Only **4 titles (8%)** came from outside the top 3 — and three of those four were barely outside it (FR 4–5).

Only 1 of 50 champions in ten years came from truly outside the expected zone: Leicester City at FR = 11.

Every Champion in Ten Years, Ranked by How Much They Beat the Model



Every champion overperformed — winning always required beating the financial prediction, not just meeting it. For most (Bayern, Real Madrid, PSG, Man City) the margin is modest: 1–3 positions. Leicester City sits alone, far left of every other point on this chart.

Robustness Checks: The Core Finding Survives Every Test

Test	Statistic	p -value	Verdict
Quadratic vs. linear (FR^2)	$F = 31.73$, 1 df	< 0.001	FR² confirmed
FR $\times$ League interaction	$F = 1.20$, 4 df	0.31	Effect uniform
FR $\times$ Season time trend	$F = 0.42$, 1 df	0.52	Stable 2015–25
Cubic term (FR^3)	$F = 0.18$, 1 df	0.67	Model H sufficient
H4 vs. H (League + Season)	$F(5) = 0.253$	0.939	Controls add nothing
Heteroskedasticity (BP test)	$\chi^2 = 45.5$	< 0.001	Present; HC1 SEs used

Models H2, H3, H4 all score higher AIC than Model H. Neither league fixed effects nor a time trend add explanatory power. Model H ($FR + FR^2$) is the correct and sufficient specification.

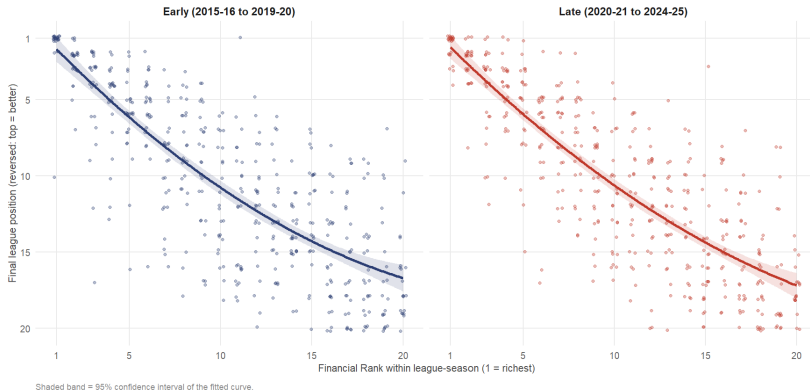
Statistical: Under HC1 robust SEs, the FR coefficient remains $p < 0.001$.

Plain English: The finding survives every specification test applied.

Has the Relationship Changed? Early vs Late Decade

Has the Money–Position Relationship Changed Over Time?

Model H (quadratic) fit separately • Early N = 490, Late N = 486



Splitting the decade at 2020 — pre- and post-COVID — the quadratic curve is visually almost identical in both halves. Early $R^2 = 0.646$, Late $R^2 = 0.698$; slopes 1.250 vs 1.210. Financial dominance did not weaken or strengthen across the decade.

The Financial Rank Slope Barely Moved Across the Decade

Financial Rank Slope: Early vs Late Period

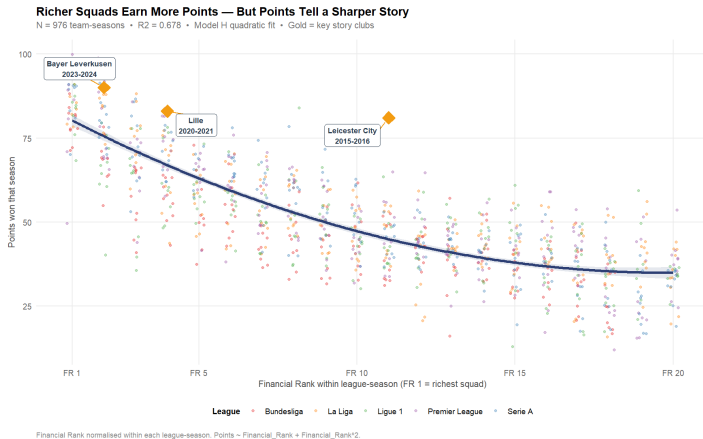
Point = OLS estimate, bars = 95% CI • Intervals OVERLAP (no strong evidence of change) • Interaction $p = 0.7966$



Model H: $\text{League_Position} \sim \text{Financial_Rank} + \text{Financial_Rank}^2$, fit separately per period.

Early slope: 1.250 [1.029, 1.470] Late slope: 1.210 [1.006, 1.415]. The 95 % confidence intervals overlap almost completely and the interaction $p = 0.80$ — no statistically meaningful change across the decade.

Robust to Outcome Measure: Points Earned Tells the Same Story

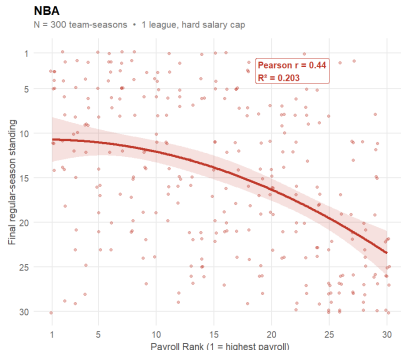
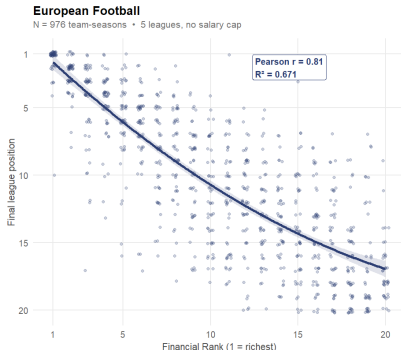


Replacing league position with points earned as the outcome variable: $R^2 = 0.678$ vs 0.671 using position — virtually identical explanatory power. Leicester City 2015–16 remains the single largest residual in the entire dataset whichever outcome measure is used.

What If Money Were Capped? Football vs the NBA

Does Money Buy Success? Football vs. the Salary-Capped NBA

Same quadratic specification (Model H) fit separately on each sport — tighter scatter and higher R^2 means money predicts finish more strongly



Football: $\text{Financial_Rank} + \text{Financial_Rank}^2$ predicting League_Position (2015-16 to 2024-25). NBA: $\text{Payroll_Rank} + \text{Payroll_Rank}^2$ predicting Final_Standing (regular-season, pooled across conferences, same seasons).

Same quadratic model, two sports. Football (no cap): $r = 0.81$, $R^2 = 0.671$, 92 % of titles to financial top 3. NBA (hard salary cap): $r = 0.44$, $R^2 = 0.203$, only 50 % of titles to top-3 spenders. A salary cap visibly and substantially weakens the money–success link. This is the counterfactual to UEFA Financial Fair Play.

Conclusions: What Ten Years of Data Tell Us

Finding 1 — Money very strongly predicts where clubs finish.

$r = 0.813$, $R^2 = 0.671$ across 976 team-seasons ($p < 0.001$): **67 %** of positional variation explained by FR alone.

Finding 2 — The relationship is non-linear: diminishing returns.

FR² term: $F = 31.73$, $p < 0.001$. Marginal penalty: **1.19 positions** per rank at the top vs. **0.62 positions** at the bottom. The gap between richest and second-richest is much larger than the gap between 15th and 16th richest.

Finding 3 — The effect is stable across leagues and over time.

FR × League: $F = 1.20$, $p = 0.31$ (uniform across all 5 leagues). FR × Season: $F = 0.42$, $p = 0.52$ (stable 2015–16 to 2024–25). Financial dominance is **universal** and **stable**, not a recent trend or league-specific.

Finding 4 — Exceptions exist, but only one is truly extraordinary.

4 of 50 champions came from $FR > 3$. Napoli (FR5), Lille (FR4), Liverpool (FR4) were within 1 SD of expectation. **Only Leicester City 2015–16** — FR=11, predicted 11th, won the title — is a genuine statistical outlier (−3.2SD, highest Cook's D in dataset).

Financial dominance is universal, stable, and statistically powerful across all five leagues and all ten seasons. Leicester City 2015–16 is the exception that proves the rule.

Conclusions: Three Additional Checks, Same Answer

Check 5 — The finding is stable across time (Early vs Late decade).

Model H re-fit independently on 2015–16 to 2019–20 ($R^2 = 0.646$) and 2020–21 to 2024–25 ($R^2 = 0.698$). Slopes 1.250 vs 1.210; 95 % confidence intervals overlap almost completely; interaction $p = 0.80$. Pre- and post-COVID: no meaningful change.

Check 6 — The finding holds regardless of how we measure success.

Replacing league position with points earned as the outcome: $R^2 = 0.678$ vs 0.671 — virtually identical. **Leicester City 2015–16 remains the single largest residual in the entire dataset** whichever outcome measure is used. Position or points, the story does not change.

Check 7 — A salary cap substantially weakens the money–success link.

Same quadratic model applied to the NBA (hard salary cap, 2015–16 to 2024–25): $r = 0.44$, $R^2 = 0.203$ vs football's $r = 0.81$, $R^2 = 0.671$. Only 50 % of NBA titles went to top-3 spenders vs 92 % in football. 8 different NBA champions in 8 years would be unthinkable in European football. This is direct evidence that **UEFA Financial Fair Play has a meaningful theoretical basis**.

Across seven independent checks — quadratic shape, league uniformity, time stability, cubic rejection, robust standard errors, alternative outcome measure, and cross-sport comparison — the core finding survives every single one. Financial dominance in European football is real, stable, and structurally built in.

Money and Success in Football

Does Financial Strength Predict League Position?

$$r = 0.813 \quad R^2 = 0.671 \quad N = 976$$

Five leagues · Ten seasons · One answer: **yes, and very strongly so**

Key Numbers to Remember

- **76 %** of titles won by FR = 1
- **92 %** of titles from financial top 3
- **67 %** of position variance explained by FR alone
- **1** true outlier in 10 years: Leicester City 2015–16

The One True Exception

Leicester City 2015–16

FR = 11 Predicted: 11th

Actual: **1st (Champion)**

Residual = -10.5 pos = -3.2 SD

The model said impossible.

They proved it — once in a generation.